Engineering Physics

Unit- IV Electron Ballistics and Quantum Mechanics

Syllabus
Motion of a charged particle in uniform electric and magnetic field, Cross field configuration, Bainbridge mass spectrograph.
Wave-particle duality, Wave packet, Heisenberg's uncertainty principle, Schröedinger's time dependent and independent wave equations, physical significance of wave function.

<u>Motion of electron in Parallel Electric field</u>

Consider an Electric field produced between two parallel plates separated by a small distance d and by applying a potential difference V across the plates such that the direction of electric field is in X- direction. Then Electric field intensity is given by

$$E = V/d$$

Let an electron is placed at rest at the negative plate. The electron will experience attraction force towards the positive plate given by

$$F = eE \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

But by Newton's Second law of Motion $\quad F = ma \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$

Therefore from (1) and (2) $\quad ma = eE$

Or acceleration of electron $\quad a = \frac{eE}{m}$

And velocity of electron $\quad v = at \quad$ or $\quad v = \frac{eE}{m}t$

And displacement $\quad x = \frac{1}{2}at^2 \quad$ or $\quad x = \frac{1}{2}\frac{eE}{m}t^2$

Thus the electron follows a straight line path in parallel electric field.
The energy gained by electron in Electric field due to applied potential V is given by

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}m\,(2\,a\,x)$$

$$or\ \ KE = \frac{1}{2}m\left(2\frac{eE}{m}\,x\right)$$

$$or\ KE = e\,E\,x = eV$$

KE for electrons is measured in terms of (eV = Electron Volt)

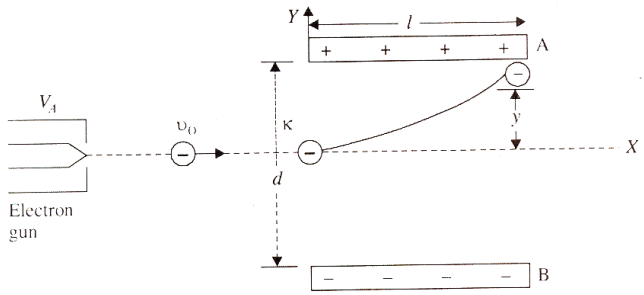
Or $\quad \frac{1}{2}mv^2 = eV$

Or velocity of electron in Electric field is given by $\quad v = \sqrt{\frac{2eV}{m}}$

Motion of electron in Uniform perpendicular Electric field

Consider an Electric field produced between two parallel plates of length l oriented horizontally and separated by a small distance d and by applying a potential difference V across the plates. The electric field is directed vertically downward in Y-direction. Then Electric field intensity is given by

$$E = V/d$$



Let an electron moving in x-direction with velocity υ_o enters the electric field. The electron experiences an upward force and gets deflected upward in y-direction towards the positive plate.

Then the force experienced by the electron

$F_x = 0$ $\quad$ $F_y = eE$

And acceleration $\quad$ $a_x = 0$ $\quad$ $a_y = \frac{eE}{m}$

And velocity $\quad$ $v_x = v_0$ $\quad$ $v_y = a_y t = \frac{eE}{m} t$

And displacement $\quad$ $x = v_0 t$ $\quad$ $y = \frac{1}{2} a_y t^2 = \frac{1}{2}\frac{eE}{m} t^2$

Eliminating t from y by using $x = v_0 t$

$$y = \frac{1}{2}\frac{eE}{m}\frac{x^2}{v^2{}_0}$$

or $\quad$ $y = kx^2$ $\quad$ where $k = \frac{1}{2}\frac{eE}{mv^2{}_0}$ is a constant

This is an equation of a parabola. It means that *electron follows a parabolic path when it enters a transverse electric field.*

Motion of electron in longitudinal Magnetic Field

When a charge particle moving with velocity **v** enters in a magnetic field **B** it experience a force called as Lorentz force F_L given by

$$F_L = qvB\, sin\theta$$

For longitudinal (parallel) magnetic field $\theta = 0$ or $\theta = 180^o$ $\quad$ $sin\theta = 0$

Therefore Lorentz force $\quad$ $F_L = 0$

Hence the magnetic force is zero. Therefore the motion of charged particle in parallel magnetic field remains unaffected

Motion of electron in perpendicular Magnetic Field

When an electron enter with a uniform velocity **v** perpendicular to a magnetic field **B** then the force due to magnetic field on electron is given by

$$F_L = evB\, sin\theta$$

The magnetic field **B** is acting into the page indicated by the crosses

Since the electron enters in transverse direction to **B** $\theta = 90°$ $\quad$ Therefore $sin\theta = 1$

And hence $\quad$ $F_L = evB$

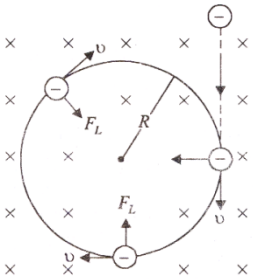
Since e, v, and B are constant the magnitude of this force F_L remains constant. Hence this force can change only the direction of motion of electron acting in perpendicular direction and deflects the electron continuously along a Circular path. Thus the force F_L act as a centripetal

That is $$F_L = \frac{mv^2}{R}$$

Or $$e\,v\,B = \frac{mv^2}{R}$$

Or $$R = \frac{mv}{e\,B}$$

Where **R** is the radius of circular path followed by the electron



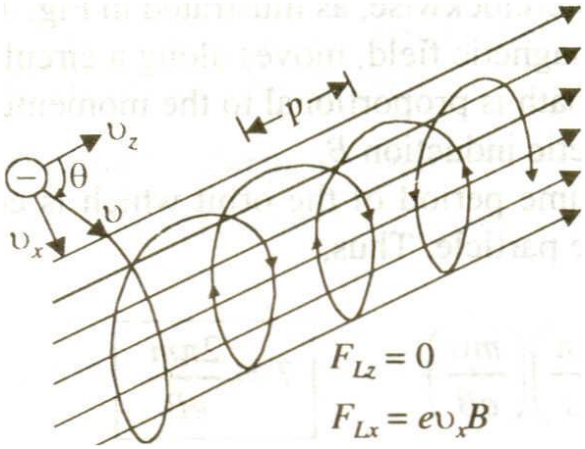
If **T** is the Time Period of electron in one revolution then

$$T = \frac{circumference}{v} = \frac{2\pi R}{v}$$

$$T = \frac{2\pi}{v}\left(\frac{mv}{e\,B}\right) = \frac{2\pi m}{e\,B}$$

$$T = \frac{2\pi m}{e\,B}$$

<u>Motion of Electron at an angle to the Magnetic field</u>



When an electron enters a uniform magnetic field **B** acting in z-direction with a velocity **υ** making an angle **θ** with the magnetic field direction then the velocity **υ** is resolved into parallel and perpendicular components

$$v_z = v\cos\theta \quad \text{and} \quad v_y = v\sin\theta$$

And the force will be $$F_z = e\,v_z\,B = 0 \quad \text{and} \quad F_y = e\,v_y\,B$$

The force F_z on the electron in the direction parallel to the magnetic field is zero. Hence electron continues to move along the field direction

The force F_y on the electron due to the magnetic field in a direction perpendicular to the field makes the electron to move in circular path. Therefore

$$F_y = \frac{mv_y^2}{R}$$

Or ` $$e\,v_y\,B = \frac{mv_y^2}{R}$$

$$e\,v\sin\theta\;B = \frac{m\,(v\sin\theta)^2}{R}$$

Or $$R = \frac{mv\sin\theta}{e\,B}$$

Thus the electron is subjected to two forces simultaneously one in parallel direction which makes the electron to move in direction of the field and the other in perpendicular direction which makes the electron to move in circular path hence it follows a Helical path

Then Pitch P of the Helix is defined as parallel distance travelled by the electron in one revolution

$$Pitch = v_z \times T$$

Or $$Pitch = v\cos\theta \times \frac{2\pi m}{e\,B}$$

<u>Motion of Electron in Crossed field</u>

When Electric and Magnetic fields act over the same region perpendicular to each other than they are said to be in **cross field configuration**.
When electron passes through the cross field configuration, in a direction perpendicular to both Electric field and Magnetic field it is deflected simultaneously by both the field.
The electric field deflects the electron in upward direction with a force $F_E = eE$
while the magnetic field deflects the electron in downward direction with a force $F_L = evB$
If the magnitudes of electric field and magnetic fields are adjusted such that force due to electric field becomes equal to force due to magnetic field than

$$F_E = F_L$$

$$eE = evB \quad \text{or} \quad v = \frac{E}{B}$$

Then the electron is not deflected and continues to move along the straight line.

<u>Velocity Filter Or Velocity Selector</u>

The velocity filter is a device, which uses electric and magnetic field in crossed configuration for selecting a stream of charged particles of single velocity from a beam of charged particles having a wide range of velocities.

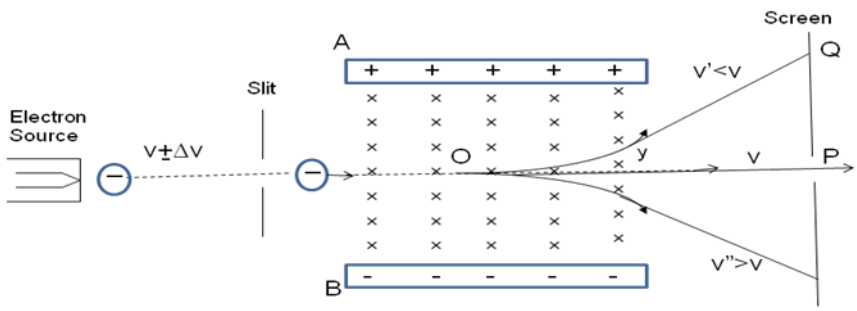


Fig 10: Velocity Filter

Suppose a beam of Electron having velocities $v \pm \Delta v$ enter the crossed field configuration. The electric field E is produced in the vertical direction by a set of charged parallel plate and the uniform magnetic field B is applied perpendicular to it (acting into the page).

The electric field deflects the electron in upward direction with a force $F_E = eE$
while the magnetic field deflects the electron in downward direction with a force $F_L = evB$

If the fields are adjusted such that the electric force balances the magnetic force acting on the electrons moving with velocity v, then

$$F_E = F_L$$

$$eE = evB \qquad \text{or} \qquad v = \frac{E}{B}$$

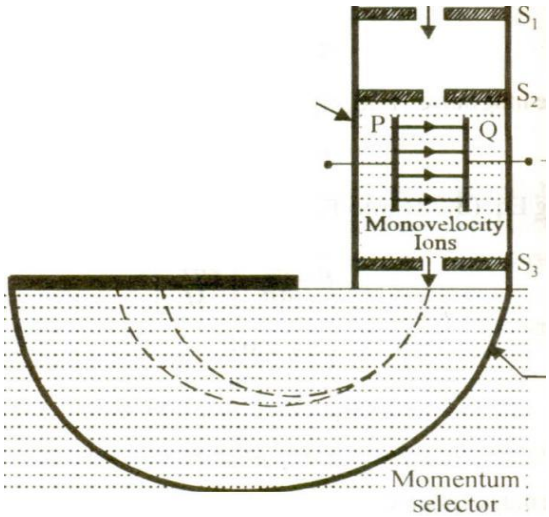
The electrons moving with velocity v, are not deflected and continue to travel along a straight path subsequently they pass through the slit at P.

Electrons moving with a lesser velocity $(v - \Delta v)$ will get deflected upward along OQ and those moving with greater velocity $(v + \Delta v)$ will get deflected downward along OR. The electrons deflected away are absorbed by the slit walls.

Thus a strictly homogeneous single velocity electrons beam travelling along OP is obtained with the help of crossed fields. This arrangement is therefore known as a velocity filter or a velocity selector.

Bainbridge Mass Spectrograph

A Mass Spectrograph is a device used to separate isotopes of an element.



Principle:
If a beam of ions having the same velocity are allowed to pass through a perpendicular magnetic field, the magnetic field deflects the ions of different masses into separate circular path.

Construction:
- It consists of two slits S_1 and S_2 which collimates the ion beam towards a velocity selector
- The velocity selector selects the monovelocity ion beam which pass through another slit S_3 arrange to further collimate the narrow monovelocity ion beam towards the analyzing vacuum chamber.
- The analysing chamber is placed in a uniform magnetic field acting perpendicular to it.
- A photographic plate is mounted in the analyzing chamber in line with the slit S_3

Working:
- Element under study is taken in the form of ions. The ion beam enters into the mass spectrograph through two slits S_1 and S_2 which collimates the ion beam towards a velocity selector
- The velocity filter consists of electric and magnetic field in crossed configuration applied normal to the direction of motion of the ions. The fields are adjusted such that the electric force balances the magnetic force acting on the ions than the ions having velocity v = E/B, emerge out of the velocity filter without any deflection and pass through the slit S_3. The ions having velocities different from v get deflected away and are absorbed by the walls of the chamber.

- Now the beam consisting of ions having the same velocity emerge out from the velocity selector through a slit S_3 and enter into the analyzing chamber where a magnetic field B is applied perpendicular to the path of the ion beam. Due to this transverse magnetic field the ions experience a magnetic force qvB and moves along a circular path

Magnetic force = Centripetal force

$$q\,v\,B = \frac{mv^2}{R}$$

or $$R = \frac{mv}{q\,B}$$

we have $$v = \frac{E}{B}$$

Therefore $$R = \frac{mE}{q\,B^2}$$

As E and B are constant $$R \propto m$$

That is radius of the circular path is directly proportional to the mass of ions

- Hence the ions having same charges but different masses travel in different circular path of different radius and produces lines on the photographic plate at different positions.

- The mass spectrum contains lines whose number gives us the number of isotopes in the given element.
- The distance of any line on photographic plate from slit S_3 is diameter of the corresponding circular path. Thus if $\boldsymbol{x}$ is the distance of line from S_3 then

$$x = 2R$$

$$x = \frac{2mE}{qB^2}$$

$$m = \frac{2E}{qB^2}x$$

Since E, B and q are known mass of ion can be determined

<u>De-Broglie Hypothesis</u>

The light shows dual nature of wave as well as particle. de-Broglie suggested that the wave-particle dualism need not be special feature of light alone but the material particles must also exhibit such dual behavior. This is known as de Broglie hypothesis.

According to de Broglie hypothesis any moving particle is associated with a wave. The waves associated with moving particles are known as de Broglie waves or matter waves.

<u>De-Broglie Wavelength</u>

Energy of a photon is given by $E = h\nu$

It can also be represented as $E = mc^2$

Therefore $h\nu = mc^2$

Momentum of photon $$p = \mathrm{mc} = \frac{\mathrm{mc}^2}{\mathrm{c}} = \frac{h\nu}{c} = \frac{h}{\lambda}$$

$$or \quad p = \frac{h}{\lambda} \qquad or \qquad \lambda = \frac{h}{p}$$

Thus the wavelength λ of matter waves associated with a particle of mass *m*, moving with velocity *v* and carrying a momentum (*mv)* is given by:

$$\lambda = \frac{h}{p} \quad or \quad \lambda = \frac{h}{mv}$$

This is called as de-Broglie wavelength.

<u>De-Broglie Wavelength associated with an accelerated charge particle</u>

If an electron is accelerated by a potential difference of V volts than its kinetic energy is given by

K.E = eV

But K.E = ½ mv^2

Or ½ $mv^2 = eV$

Or $v = \sqrt{\frac{2eV}{m}}$

Therefore the de Broglie wavelength is given by $\lambda = \frac{h}{mv}$

Or $\lambda = \frac{h}{m\sqrt{\frac{2eV}{m}}} = \frac{h}{\sqrt{2meV}}$

$$\lambda = \frac{h}{\sqrt{2meV}}$$

Substituting the values λ for electron is

$$\lambda = \frac{6.623 \times 10^{-34}}{\sqrt{2\times 9.1\times 10^{-31}\times 1.6\times 10^{-19}}} = \frac{12.26\times 10^{-10}}{\sqrt{V}} = \frac{12.26\ \text{A}^{\circ}}{\sqrt{V}}$$

E = eV Therefore λ can also be written as

$$\lambda = \frac{h}{\sqrt{2mE}}$$

<u>Properties of matter waves</u>

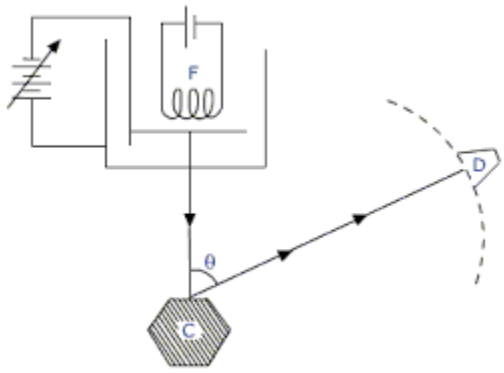
The wavelength of matter waves is given by $\lambda = \frac{h}{mv}$

1. $\lambda \propto \frac{1}{m}$, hence lighter the particle greater is the wavelength associated with it.
2. $\lambda \propto \frac{1}{v}$, hence greater the velocity of the particle smaller is the wavelength associated with it.
3. Matter waves are produced by the motion of the particles and are independent of the charge. Therefore they are not electromagnetic wave.
4. They travel through vacuum and do not require material medium for their propagation.
5. They do not have constant velocity. Their velocity is equal to the velocity of particle.
6. They exhibit diffraction phenomenon as any other waves.

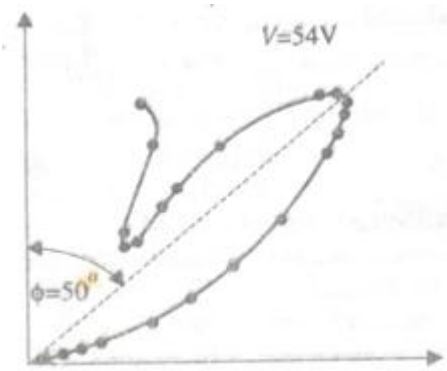
<u>Davisson-Germer experiment</u>
<u>Experimental verification of de-Broglie wavelength</u>

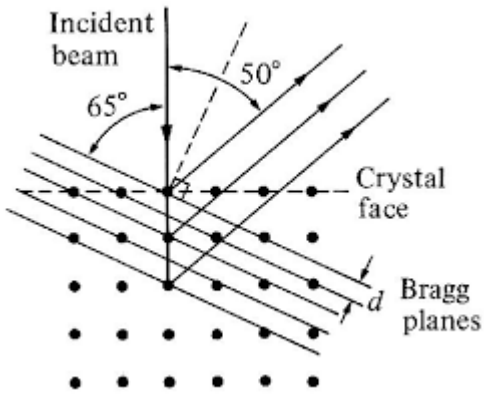
- Waves exhibit diffraction. If the de-Broglie hypothesis is valid, then the matter waves should also exhibit diffraction effects. In 1927, Davisson and Germer observed the diffraction of an electron beam incident on a nickel crystal. This experiment provided a convincing proof of wave nature of matter (electron).

- The experimental set up of Davisson and Germer is shown in fig. It consists of an electron gun to obtain a fine narrow beam of electron. The electron beam is allowed to fall normally on a single crystal of nickel. These electrons are scattered by nickel crystal. The number of electrons scattered by the crystal in different directions is studied with the help of a detector, which can be moved on a circular scale.



- Classical wave theory predicts that the scattered electron will be detected in all direction. But Davisson and Germer found that the number of electron scattered in any direction depends on the scattering angle θ.
- A polar graph was then plotted between the detector current and the angle between the incident beam and the diffracted beam. According to classical wave theory if electron scattered uniformly in all direction the graph would have been a circle. But the intensity of the scattered electron beam in the graph exhibits distinct maxima and minima. Electrons scattered more pronouncedly (maxima) at an angle of 50º for a potential difference of 54 volts.





- The maximum is an indication that electrons have wave nature and electrons are being diffracted at the atomic planes (rows of atoms) in the crystal. Bragg's law, applicable for x-ray diffraction by crystals, would be valid for electron wave diffraction also and can be used to calculate the wavelength of electrons.
- Fig shows the Bragg's diffraction of electron at the atomic planes of the nickel crystal. Both the incident electron beams and the scattered electron beams makes an angle of 65º with the atomic planes and the inter-planer spacing of the atomic planes of the Nickel crystal is d = 0.91Å

  Taking n =1 for first order maximum, the Bragg's law is

  $$n\lambda = 2d \sin\theta$$

  $$\text{or } \lambda = 2d \sin\theta = 2\times0.91\times\sin 65^{o} = 1.65Å$$

- The wavelength associated with electron accelerated by the potential of 54 V calculated by using de Broglie equation is given by

$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{6.62\times10^{-34}J.s}{\sqrt{2\times9.11^{-31}kg\times1.602\times10^{-19}C\times54V}} = 1.66Å$$

- The agreement between the two values obtained from different considerations is excellent proof for the wavelike character of electrons and therefore for the de Broglie hypothesis.

<u>Heisenberg's Uncertainty Principle</u>

Heisenberg's Uncertainty principal states that "It is impossible to measure simultaneously the exact position and momentum of a particle to an unlimited accuracy."

If Δx is the uncertainty in position of the particle and Δp_x is the uncertainty in momentum of particle, then, the product of uncertainty in position and uncertainty in momentum is greater than or equal to $h/2\pi$

$$\Delta x . \Delta p_x \geq h/2\pi$$

<u>Physical significance of Heisenberg's Uncertainty Principle</u>

- We cannot determine the exact position or accurate momentum of a particle
- But we can only determine the probability of finding the particle at a certain position
- Or the probable value of the momentum of a of a particle
- These probabilities are obtained by probability density function

<u>Application of Heisenberg's Uncertainty principle:-</u>

Uncertainty in Energy and Time:

Kinetic energy is given by $E = \frac{1}{2} m v^2$

Then uncertainty in energy $\Delta E = \frac{1}{2} mv \times 2\,\Delta v = v\,(m\,\Delta v)$

But $(m\,\Delta v) = \Delta p$

Therefore $\Delta E = v\,\Delta p$

But again $v = \frac{\Delta x}{\Delta t}$

Therefore $\Delta E = \frac{\Delta x}{\Delta t}\,\Delta p$

Or $\Delta t\,\Delta E = \Delta x\,\Delta p$

But by Heisenberg's uncertainty principle $\Delta x . \Delta p_x \geq h/2\pi$

Therefore $\Delta t\,\Delta E \geq h/2\pi$

From the above equation it is seen that the product of uncertainties in Energy and Time measurement is equal to the inequality of $h/2\pi$ hence there is uncertainty in time and energy

Electrons cannot exist in the nucleus:

- The radius of the nucleus is of the order of 10^{-14} *m* .Therefore, if electron exists inside the nucleus, the maximum uncertainty Δx in the position of the electron is equal to the diameter of the nucleus. Thus, $\Delta x = 2 \times 10^{-14}$ *m*
- The minimum uncertainty in momentum is then given by

$$\Delta p = \frac{h}{2\pi\Delta x} = \frac{1.05 \times 10^{-34}\, Js}{2 \times 10^{-14}} = 5.2 \times 10^{-21} Kg\, m/s$$

- The minimum uncertainty in momentum can be taken as the momentum of the electron. Thus,
  $p = 5.2 \times 10^{-21} Kg\, m/s$
- The minimum energy of the electron in the nucleus is given by
  $E_{min} = mc^2 = pc = (5.2 \times 10^{-21} Kg\, m/s) \times (3 \times 10^8\, m/s)$
  $= 1.56 \times 10^{-12} J = 9.7$ MeV
- It implies that if an electron exists within the nucleus, it must have a minimum energy of about 10 MeV. But the experimental measurements showed that the maximum kinetic energies of electron is of the order of 4 MeV only. Hence electrons cannot be present in the nucleus.

<u>Significance of wave function, 'ψ'</u>

- A wave function ψ describes the motion of an electron.
- The wave function ψ gives the probability of finding a particle at any specific point (x y z) at time t.
- The wave function ψ is complex quantity but the probability of finding the particle must be real. Therefore to make probability a real quantity, ψ must be multiplied by its complex conjugate ψ* i.e. ψψ*.
- Thus $|\psi|^2$ is called as the probability density.
- The probability of finding the particle in infinitesimal small volume dV = (dx dy dz) is given by
  $P = |\psi(x, y, z)|^2\, dx.dy.dz$

  $P = |\psi|^2\, dv$

- The probability P should be equal to 1 because particle is definitely somewhere in space and hence the integral of $|\psi|^2$ dV over the entire space must be equal to 1. That is
  $\iiint_{+\infty}^{-\infty} |\psi|2\ dx\, dy\, dz = 1$

  This is called as Normalized condition

<u>The Schrodinger time independent wave equation (STIWE)</u>

The differential wave equation of De-Broglie wave is given by

$$\frac{\partial^2\psi}{\partial t^2} = v^2 \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2}$$

This can be written as $\frac{\partial^2\psi}{\partial t^2} = v^2\, \nabla^2\psi$ ..... (1)

Where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the del operator and ψ is a wave function

The solution of this differential wave equation is of the form

$$\psi = \psi_0 \sin \omega t$$

Where $\omega = 2\pi\nu$ Therefore $\psi = \psi_0 \sin\ (2\pi\nu)t$ ..... (2)

Differentiating equation (2) $\frac{\partial\psi}{\partial t} = \psi_0 \cos 2\pi\nu t \times (2\pi\nu)$

Again differentiating $\frac{\partial^2\psi}{\partial^2 t} = -\psi_0 \sin 2\pi\nu t \times (2\pi\nu)^2$

Or $\frac{\partial^2\psi}{\partial^2 t} = -4\pi^2\nu^2\, \psi_0 \sin 2\pi\nu$

Using equation (2) $\frac{\partial^2\psi}{\partial^2 t} = -4\pi^2\nu^2\,\psi$

Substituting this value in equation (1) $-4\pi^2\nu^2\,\psi = v^2\,\nabla^2\psi$

But we have $v = \nu\,\lambda$ therefore $-4\pi^2\frac{v^2}{\lambda^2}\,\psi = v^2\,\nabla^2\psi$

Or $-\frac{4\pi^2}{\lambda^2}\psi = \nabla^2\psi$

Or $\nabla^2\psi + \frac{4\pi^2}{\lambda^2}\psi = 0$ ..... (3)

Using De-Broglie hypothesis $\lambda = \frac{h}{mv}$ in equation (3)

We get $\nabla^2\psi + \frac{4\pi^2 m^2 v^2}{h^2}\psi = 0$ ..... (4)

The total energy E of the particle in De-Broglie wave is given by

Total Energy = Kinetic Energy + Potential Energy

Or $E = \frac{1}{2}mv^2 + V$

Or $\frac{1}{2}mv^2 = E - V$

Or $m^2v^2 = 2m(E - V)$

Substituting this value in Equation (4)

$$\nabla^2\psi + \frac{4\pi^2 2m(E-V)}{h^2}\psi = 0$$

Or $\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$ ..... (5)

This is called as Schrodinger Time Independent Wave Equation

Also we have $\hbar = \frac{h}{2\pi}$ therefore we can write the equation as

$$\nabla^2\psi + \frac{2m}{\hbar^2}(E-V)\psi = 0$$

<u>The Schrodinger time dependent wave equation (STDWE)</u>

For the Schrodinger time dependent wave equation the differential equation

$$\frac{\partial^2\psi}{\partial t^2} = v^2\,\nabla^2\psi$$

has the solution of the form $\psi = \psi_0 \, e^{-i\omega t}$ …….. (6)

Differentiating the above equation $\frac{\partial \psi}{\partial t} = \psi_0 \, e^{-i\omega t} \times (-i\omega)$

Or $\frac{\partial \psi}{\partial t} = -i\omega \, \psi_0 \, e^{-i\omega t}$

Or $\frac{\partial \psi}{\partial t} = -i\omega \, \psi$

Now we know that $\omega = 2\pi\nu$ therefore $\frac{\partial \psi}{\partial t} = -i(2\pi\nu) \, \psi$

and $E = h\nu$ therefore $\nu = \frac{E}{h}$

Substituting this value in above equation we get

$$\frac{\partial \psi}{\partial t} = -i \, 2\pi \frac{E}{h} \, \psi$$

Or $\frac{\partial \psi}{\partial t} = -i \, \frac{E}{\hbar} \, \psi$ because $\hbar = \frac{h}{2\pi}$

Therefore $-\frac{\hbar}{i} \frac{\partial \psi}{\partial t} = E \, \psi$

Substituting this value of $E \, \psi$ in Schrodinger Time Independent Wave Equation

$$\nabla^2 \psi + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

or $\frac{\hbar^2}{2m} \nabla^2 \psi + (E - V)\psi = 0$

$$\frac{\hbar^2}{2m} \nabla^2 \psi - \frac{\hbar}{i} \frac{\partial \psi}{\partial t} - V\psi = 0$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi = \frac{\hbar}{i} \frac{\partial \psi}{\partial t}$$

This is called as Schrodinger Time dependent Wave Equation